

The Spatial-Pressure Gravitational Theory (SGT): Flagship Manifesto and Milestone Snapshot

——Theoretical Architecture, Core Theorems, Model Predictions, and Engineering Outlook

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Abstract

Serving as the flagship manifesto and milestone snapshot of the Spatial-Pressure Gravitational Theory (SGT), this paper systematically consolidates the phased achievements of the theory from a conceptual model to a self-consistent framework. SGT postulates that the vacuum is filled with a continuous elastic medium—the "spatial-energy medium"—and that gravity originates from the tension gradient generated when the medium is stretched

by matter. This paper strictly delineates the **Proven Theorems**, **Current Model Predictions**, and **Engineering Outlook** of the theory. We have rigorously proven the Non-Emptiability Theorems and the Cosmological Singularity Elimination Equivalence Theorem. Under the current parameterized framework, the numerical solver yields specific predictions such as $r_{\text{ISCO}} = 4.23M$ and a gravitational wave echo delay of $\sim 14.3M$. Furthermore, we propose a future roadmap for building the SGT Solver computational infrastructure. This paper aims to provide a unified terminology map and citation hub for the SGT framework, laying the foundation for subsequent specialized papers.

Keywords: Spatial-Pressure Gravitational Theory; Medium Mechanics Ontology; Singularity Resolution; Non-Emptiability; Gravitational Wave Echoes; Anisotropic Stress

The Three Pillars of SGT

Pillar I: Global Mathematical Closure

One set of equations, two independent parameters, covering all gravitational scenarios from galactic rotation to cosmic explosion.

Unlike mainstream modified gravity theories that rely on artificial "matching" or external scalar fields across different scales, the multi-scale behavior of SGT is an **intrinsic emergence** of the same set of nonlinear medium elastic equations. The weak-field limit naturally recovers MOND

behavior, the medium-field regime precisely restores General Relativity (GR), the strong-field region automatically triggers gravity shielding ($\chi \rightarrow 0$), and cosmological evolution naturally leads to singularity resolution. The mathematical boundaries of SGT are not "spliced on"; they "grow out" of the medium ontology.

Pillar II: Singularity Resolution Theorem

Singularities are markers of theoretical failure; SGT is the only theory that provides finite answers at the point of failure.

GR predicts infinite singularities at the center of black holes and the Big Bang. SGT physically truncates singularities at the classical level through the **Non-Emptiability** principle: in black holes, the divergence of the radial elastic modulus locks the stretching parameter $f \leq 1$; in cosmology, the equivalence theorem rigorously proves $f(0) = 1 \Leftrightarrow H(0) < \infty$. While Loop Quantum Gravity and String Theory have struggled to achieve global singularity resolution after decades, SGT accomplishes this unified goal for both black holes and cosmology relying purely on a classical continuous medium mechanics framework.

Pillar III: Falsifiable Predictions

SGT provides highly falsifiable experimental predictions, not adjustable parameter fittings.

Core predictions include $r_{\text{ISCO}} = 4.23M$ (GR: $6.00M$), gravitational wave

echo delay $\sim 14.3M$ (GR: no echoes), and CMB scalar spectrum low- ℓ suppression. These are hard mathematical outputs of the field equations, not tunable fits. Every prediction has clear falsifiable boundaries and will be independently tested by next-generation observatories: Athena (ISCO shift), 3G GW detectors (echoes), and CMB-S4 (low- ℓ suppression).

I. Physical Picture and Ontological Foundation

SGT does not modify the mathematical formalism of GR; rather, it provides a physical interpretation based on an **elastic medium ontology**.

- **Static Gravity:** Matter acts as a local topological defect, stretching the surrounding medium. Gravity is essentially a "push" generated by the tension gradient, not an action-at-a-distance "pull."
- **Dynamic Gravitational Waves:** Gravitational waves are the escape of elastic waves when the strain gradient is forced to detach from the source due to the medium's inability to synchronously respond to violent perturbations.
- **The Nature of the Metric:** The spacetime metric is not the physical ontology, but a mathematical translation layer of the medium's stretching state (f).

II. Layer 1: Proven Theorems and Core Architecture

2.1 The Non-Emptiability Theorems

The principle that the medium cannot be completely evacuated ($f \leq 1$) is the core physical constraint.

- **Static Analytical Proof:** Rigorously proven (Grade A) that as $\chi \rightarrow 0$, the radial elastic modulus C_{rr} diverges, mathematically locking $f \leq 1$.
- **Dynamic Conservation Proof:** Proven via the positive definiteness of the energy functional that $\chi > 0$ strictly holds throughout time evolution.
- **Equivalent Proposition:** $C_{rr} \rightarrow \infty$ and Non-Emptiability are strictly equivalent, constituting the SGT version of the "Cosmic Censorship Hypothesis."

2.2 Cosmological Singularity Elimination Equivalence Theorem

In the FRW background, we strictly derived: $f(0) = 1 \Leftrightarrow \chi(0) = 0 \Leftrightarrow H(0) < \infty$. The Big Bang singularity is physically truncated.

III. Layer 2: Current Model Predictions

3.1 Black Hole Structure and Thermodynamics

Based on the spherically symmetric static vacuum numerical solution:

- **Event Horizon:** $r_H = 1.6673M$ (GR: $2M$).

- **Photon Sphere:** $r_{\text{ph}} = 2.9954M$ (GR: $3M$).
- **ISCO:** $r_{\text{ISCO}} = 4.23M$ (GR: $6.00M$). *Mechanism:* $G_{\text{eff}} \rightarrow 0$ in the strong-field region shields gravity.
- **Hawking Temperature:** $T_H^{\text{SGT}} = 0.866 \times T_H^{\text{GR}}$.

3.2 Gravitational Wave Echoes

- **Echo Delay:** $\Delta t \approx 14.3M$.
- **Reflection Signature:** $R(\omega) \propto \omega^\alpha$ ($\alpha > 0$), high-frequency components reflect stronger.

IV. Comparison with Mainstream Quantum Gravity Theories

Dimension	Loop Quantum Gravity	String Theory	SGT
Ontology	Quantized math structure	Strings (unobservable)	Spatial-energy continuum (macroscopically correspondent)
Singularity	Partial (Quantum bounce)	Partial (Fuzzball)	Complete (Classical truncation, unified)
Predictions	Very weak / None	None	Multiple (X-ray, GW, CMB)
Parameters	Few but unverified	Vast (Landscape)	Only 2 independent (K, f_c)

V. Observational Targets

Prediction	SGT Prediction	GR Prediction	Target Observatory
ISCO Radius	4.23 M	6.00 M	Athena
GW Echo Delay	$\sim 14.3M$	None	3G GW Detectors
Big Bang Singularity	$H(0)$ is finite	∞	CMB-S4
Scalar Dipole Rad.	Exists	None	3G GW Network

VI. Layer 3: Roadmap and Engineering Outlook

- **SGT Solver:** Build a unified numerical solver. Expected to be orders of magnitude faster than GR numerical relativity tools for spherically/axisymmetric backgrounds.
- **Spatial Simulation:** Interface with 3D engines for real-time gravitational lensing rendering (GR/SGT dual-mode).
- **Aerospace Navigation:** Develop SGT multi-body orbital evolution integrator for space-based GW detector constellation design.

VII. Citation Hub and Index

- **SGT-PAPER-012** (This paper): Flagship Manifesto.
- **SGT-PAPER-014:** Elastic Constant System and Cross-Validation.
- **SGT-PAPER-015:** Spherically Symmetric Static Vacuum Numerical Solution.

- **SGT-PAPER-016:** CH Series Gravitational Structural Stability Testing Framework.
- **SGT-PAPER-017:** SGT Black Hole Physics: From Static Structure to Rotational Negative Feedback.
- **SGT-PAPER-018:** Cosmological Singularity Elimination and Very Early Universe.

VIII. Conclusion

SGT provides answers to the extreme conditions of black holes and cosmology that GR cannot. A black hole is not a bottomless pit in spacetime, but a frozen state of the medium; the Big Bang is not a mathematical singularity, but a phase transition of medium stiffness depletion.

General Relativity explains **what** gravity is using spacetime curvature;

SGT explains why gravity exists using medium tension.

In the history of physics, theories that explain "why" often go much further than those that merely describe "what." The SGT framework is now ready to be tested by the stars.

Ownership and Evidence Declaration

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